TRANSLATION IMAGE T X_K U -> G IS OPEN FOR A GROUP SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma morphisms-lemma-scheme-over-field-universally-open (canonical tag 0383). cited at proof lines 972-973. G_K -> G is open for any field extension K/k.

Source reference. Original lemma 0383: sources/stacks-chow/tags.txt, source lines 4222–4222. Cited at proof lines 972-973: G_K -> G is open for every field extension; load-bearing.

Lemma 0.2. *If (G, m) is a group scheme over a field k. Let $U \subset G$ open and $T \to G$ a morphism of schemes. Then the image of the composition $T \times_k U \to G \times_k G \to G$ is open.*

Proof. For any field extension K/k the morphism $G_K \to G$ is open (Morphisms, Lemma 0383). Every point ξ of $T \times_k U$ is the image of a morphism $(t, u) : \mathrm{Spec}(K) \to T \times_k U$ for some K. Then the image of $T_K \times_K U_K = (T \times_k U)_K \to G_K$ contains the translate $t \cdot U_K$ which is open. Combining these facts we see that the image of $T \times_k U \to G$ contains an open neighbourhood of the image of ξ. Since ξ was arbitrary we win. □

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